CCC-C-421
APRIL 12, 1961
SUPERSEDING
Fed. Spec. CCC–D–181
March 30, 1944 and
Fed. Spec. CCC–D–186
March 30, 1944

FEDERAL SPECIFICATION

CLOTH, COTTON, DENIM, SHRUNK AND UNSHRUNK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton, denim cloth.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The cloth shall be of the following types, classes, and styles as specified (see 6.2 and 6.4), and table I:

Type I.—8.04 to 8.78 ounces.
Type II.—8.95 to 9.75 ounces.
Type III.—9.85 to 10.97 ounces.
Type IV.—11.06 to 12.30 ounces.

Class 1.—Unshrunk.
Class 2.—Shrunk.

Style A.—Whiteback.
Style B.—Colored filling.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC–T–191—Textile Test Methods.
PPP–P–51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations of the Textile Fiber Products Identification Act effective March 3, 1960.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the sample for shade and shall be equal to or better than the sample with respect to all characteristics for which the sample is referenced (see 6.3).

3.2 Material. The cotton shall be thoroughly cleaned, carded, drawn and evenly spun into singles yarn.

3.3 Color.

3.3.1 The color shall be as specified and shall match the standard sample when available (see 3.1, 6.2, and 6.3). Style A, white-back cloth shall contain dyed warp yarns and white or tinted filling yarns. Style B, colored filling cloth shall contain dyed warp and filling yarns.

3.3.2 *Color matching.* The color shall match the available standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.3 *Colorfastness.* The dyed cloth shall show colorfastness to light, water, laundering, and crocking equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, or when the standard sample is not a reference for colorfastness, the dyed cloth shall show good colorfastness to light, water, laundering, and fair colorfastness to crocking when tested as specified in 4.5.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I for the applicable type and class when tested as specified in 4.5.

3.4.1 *Weave.* Unless otherwise specified, the weave shall be a 3-harness,

$$\frac{(2\ \ )}{(\ 1\ )}$$

2 up, 1 down right-hand twill. (See 6.2.)

3.4.2 *Width.* The width shall be as specified, and shall be the minimum acceptable width exclusive of selvages. (See 6.2.)

3.5 Nonfibrous material. The total nonfibrous material content of type I cloth shall

TABLE I.—*Physical requirements*

Types	Classes	Weight per square yard (minimum)[1]	Yarns per inch (minimum)		Breaking strength, grab method (minimum)	
			Warp	Filling	Warp	Filling
		Ounces			*Pounds*	*Pounds*
I	1	8.04	62	39	145	53
	2	8.78	63	42	145	58
II	1	8.95	66	43	150	65
	2	9.75	67	46	150	70
III	1	9.85	68	42	167	70
	2	10.97	69	45	167	75
IV	1	11.06	70	42	177	85
	2	12.30	71	45	177	90

[1] Includes a 2½ percent minus tolerance from nominal weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

not exceed 14 percent and for types II, III, and IV cloths shall not exceed 13 percent, when tested as specified in 4.5.

3.6 Shrinkage. Class 2 cloth shall be preshrunk and shall not shrink more than 2 percent in either the direction of the warp or filling when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth or on the ticket or package.

3.7 pH. The pH of the cloth shall be not lower than 8.0 nor more than 10.0 when tested as specified in 4.5.

3.8 Length. Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 50 yards.

3.9 Fiber identification. Each piece, roll, or bolt of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act, effective March 3, 1960.

3.10 Workmanship. The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the limits set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing to assure that where color is specified the yarn is colored prior to weaving, and that piece-dyeing is not employed. The Government reserves the right to exclude from consideration for acceptance any cloth for which in-process inspection has indicated non-conformance.

4.3 Examination of the end item. Except when otherwise specified by Civil Government agencies examination shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated. The cloth shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.3.1, 4.3.2 and 4.3.3.

4.3.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the face side. The defects listed below shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The lot size shall be expressed in units of 1 yard and the inspection level shall be level II of Military Standard MIL–STD–105 for determining sample size. The acceptable quality levels (AQL's) for this examination shall be 2.5 major and 10.0 total defects (major and minor combined) per 100 units (yards). The number of pieces from which the sample is to be selected shall be in accordance with table II herein. An approximate equal number of yards shall be examined in each piece of the sample. Not more than 1 piece shall be selected from each shipping container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

Glossary Numbers Fed. Std. No. 4	Defect	Classification	
		Major	Minor
90, 91, 92, 55, 56	Hole, cut or tear—any	X	
78	Smash	X	
11, 33	Broken or missing yarns:		
	—Two or more contiguous more than ¾ inch missing	X	
	—Single, 3 inches or more missing		X
9, 12, 35	Open place, crack, or hitchback opening at widest point two or more filling yarn diameters or sequence of openings resulting in a weak spot	X	
	Clearly visible,* but opening at widest point less than two filling yarn diameters or raised places resulting from slack warp ends		X
4, 10	Mispick—resulting in abrupt break in sequence of weave, double picks or multiple skips	X	
13, 34	Float or skips:		
	—Multiple, ½ inch or more in combined warp and filling direction	X	
	—Multiple less than ½ inch in combined warp and filling direction or single floating over 3 inches or more		X
79	Abrasion mark—resulting in a thin or weak place	X	
8, 9	Fine filling bar or light place, clearly visible*	X	
2, 26	Coarse filling bar, or heavy place—clearly visible*		X
3, 5, 27, 77	Slough off, jerked in filling, slub, slug or gout:		
	—3/32 inch or more in thickness and over 1 inch in length	X	
	—3/32 inch or more in thickness and less than 1 inch in length		X
	—Less than 3/32 inch in thickness but more than 3 inches in length		X
40, 41, 37, 38	Tight or slack end, slack or tight twist end, wrong draw or reed mark—clearly visible*		X
49, 50	Selvage tight—resulting in pronounced slackness or waviness in cloth		X
51	Scalloped selvage—deepest indentation exceeds 1½ inches		X
93	Baggy, ridgy or wavy cloth, clearly noticeable		X
(74, 75, 76)[1], 69, 70	Spot, stain or streak:		
	—Clearly visible* and extending 1 inch in combined length and width dimensions	X	
	—Clearly visible* extending less than 1 inch in combined length and width dimensions		X
14, 57	Shade bar, mixed filling—clearly visible*	X	
59, 61	Crease or wrinkle—embedded		X

* At normal inspection distance (approximately 3 feet).

[1] Except defects of spots, stains and streaks shall be counted regardless of cause, i.e., not restricted to rust, oil or grease.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.3.2 *Overall examination.* The pieces for this examination shall be those selected for yard-by-yard examination. Each defect listed below shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The number of defects acceptable shall be as shown in table II for the appropriate number of pieces in the sample.

Defects

Objectionable odor or not clean throughout.

Cloudy, mottled, or streaky throughout.

Uneven weaving throughout.

Selvage rolled, folded, corded, or slack, continuously or intermittently throughout the piece.

Off shade, shaded end to end, side to center, side to side or shaded throughout.

Width less than specified.

Preshrinking process or trademark identified on the cloth.

Filling yarn for style A not natural or tinted as specified.

TABLE II.

Lot size in yards	Sample size	Acceptance number
Up to 1300*	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 and up	25	2

* If lot contains fewer than 3 pieces, each piece in the lot shall be examined.

4.3.3 *Examination for length.* The pieces for this examination shall be those selected for yard-by-yard examination.

4.3.3.1 *Individual pieces.* Each piece shall be examined for gross length. Any gross length found to be less than specified, or any gross length found to be more than 2 yards below the gross length marked on the piece ticket, shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of defects acceptable shall be as shown in table II for the appropriate number of pieces in the sample.

4.3.3.2 *Total yardage in the sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the piece tickets.

4.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
	Preshrinking process identified by name or trademark on container.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross/net weight exceeds requirements.

4.5 Testing of the end item. The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 3½ continuous yards, full width of cloth. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified.

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method No.
Weave[1]	3.4.1	Visual
Weight	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Nonfibrous material[2]	3.5	2611
Shrinkage in laundering[3]	3.6	5550
Acidity (pH)	3.7	2811
Colorfastness:		
Light[4]	3.3.3	5660
Laundering	3.3.3	5610
Crocking	3.3.3	5651
Water	3.3.3	5630

[1] One determination shall be made from each unit of product and the result shall be reported as "pass or fail".

[2] The nonfibrous materials content shall be determined with the following formula:

$$\text{Nonfibrous-materials, content} = \frac{O - S}{O} \times 100$$

[3] The three specimens for this test shall be selected from the cloth (unit of product) as follows: one specimen from each side of the cloth within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns.

[4] In the event of a dispute resulting from the test with Method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with Method 5662.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment for civilian agency procurements.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Levels A, B, and C.* The cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Levels A, B and C.* The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of clothing items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class and style required (see 1.2).
(*c*) Color of cloth required (see 3.3.1).
(*d*) When weave required is other than specified (see 3.4.1).
(*e*) Width required (see 3.4.2).
(*f*) When minimum length is other than specified (see 3.7).
(*g*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of cloth.

6.3 Standard sample. For access to a standard sample, address the procuring office issuing the invitation for bids.

6.4 This document supersedes the following Federal Specifications:

CCC–D–186, Denim; Unshrunk, March 30, 1944

CCC–D–181, Denim; Shrunk, March 30, 1944

Specification CCC–C–421	*Supersedes*
Type I	
Class 1	Class A of CCC–D–186
Class 2	Class A–S of CCC–D–181
Type II	
Class 1	Class B of CCC–D–186
Class 2	Class B–S of CCC–D–181
Type III	
Class 1	Class C of CCC–D–186
Class 2	Class C–S of CCC–D–181
Type IV	
Class 1	Class D of CCC–D–186
Class 2	Class D–S of CCC–D–181
Style A	Type I of CCC–D–181 and CCC–D–186
Style B	Type II of CCC–D–181 and CCC–D–186

6.5 Level B packaging and packing should be used by ordering agencies, only when shipments are known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.6 Transportation description. Transportation descriptions and maximum weights applicable to this commodity are:

Rail:

Cotton piece goods, not otherwise indexed by name, in the original piece.

Carload minimum weight—any quantity.

Motor:

Cotton piece goods, not otherwise indexed, in the original piece.

Motor volume minimum weight—any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—MQ
Navy—MC
Air Force—WADD.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.